

EXCEL

classmate

Date _____

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Excel :- Microsoft Excel is a spreadsheet used to store, organize, calculate, analyze and visualize data.

Workbook :- An Excel file is called an workbook.

Worksheet :- single sheet inside a workbook.

Cell :- row and column meet is called cell

Row :- Horizontal line of cell
eg :- 1, 2, 3.

Column :- Vertical line of cell
eg :- A, B, C.

Cell Address :- location of cell
eg :- A1, B1, C2.

Range :- group of cell such as A1:A10 (or)
A1:C5.

Cell Address :-

Cell Address = Column + Row.

eg :- B5 = Column B + Row 5.

Types of data :-

Text - Harishma, Bangalore.

Number - 100, 3500, 10.

Date - 19/09/2026.

Time - 10:30 Am.

percentage - 75%.

Currency - ₹ 20000.

Editing a Cell :-

Double click (or) select F2.

Delete Data :-

Select the cell → Press Delete.

Auto Fill :-

Excel can automatically continue a pattern.

Formulas :-

* An Expression used to perform calculations in excel.

* Every formula start with =.

Example :-

$$A_1 = 500$$

$$B_2 = 200.$$

$$\text{sum} :- [= A_1 + B_1]$$

$$\text{Subtraction} :- [= A_1 - B_1]$$

$$\text{multiplication} :- [= A_1 * B_1]$$

$$\text{division} :- [= A_1 / B_1]$$

$$\text{power} :- [= A_1 ^ B_1]$$

sum, Average, Count.

i) sum() :- used to Add numbers.

$$[= \text{sum}(A_1 + B_1)] \text{ (or) } [= \text{Sum}(A_1 : A_5)]$$

For example :- A

Then write formula

10

20

30

40

50

= sum(A1:A5) in a

cell

Result :- 150.

2) Average () :- used to find Average.

Formula :- $[= \text{AVERAGE}(A1:A5)]$

Result :- 30

3) Count () :- used to count cell containing numbers.

Formula :- $[= \text{COUNT}(A1:A5)]$

Example :-

A
10
20
Apple
30
40

Result :- 4. Because it only count numbers not text.

Text function :-

* Very useful for Data Cleaning in data analytics.

1) Proper() :-

First letter of each word is capitalized.

example :-

$[= \text{PROPER}(A1)]$

A

Harishma Ravichandran

Result

B

$= \text{PROPER}(A1)$

Harishma Ravichandra

2) Len() :- Returns the number of characters in a cell

Syntax :- $[= \text{LEN}(A1)]$

Example :- $A1 = \text{Harishma}$

$\rightarrow [= \text{LEN}(A1)] \rightarrow \text{Result} :- 8$

space are also counted as characters

Syntax :- $[= \text{LEN}(\text{"Data Analyst"})]$

Result :- 10

3) LEFT() :- extract characters from left side of a text

Syntax :- $= \text{LEFT}(A1, \text{num of Characters})$

example :- $A1 = \text{Bangalore}$

$[= \text{LEFT}(A_4, 4)] \Rightarrow \underline{\text{Result}} :- \text{Bang}$

$\Downarrow$
Only 4 from left side of character is printed.

4) RIGHT() :- extracts characters from Right side of a text.

Syntax :- $[= \text{RIGHT}(A_1, \text{num of character})]$

Example :- $A_1 = \text{Bangalore}$

$[= \text{RIGHT}(A_1, 4)] \Rightarrow \underline{\text{Result}} :- \text{lore}$

5) MID() :- extracts characters from middle of a text.

Syntax :-

$[= \text{MID}(\text{text}, \text{start num}, \text{num of char})]$

Example :- $A_1 = \text{Data Analyst}$

$[= \text{MID}(A_1, 6, 6)]$

$\Downarrow$

Result :- ANALYS

only six letters is printed from sixth position.

6) UPPER() :- converts all text into capital letters

Syntax :- $[=UPPER(A1)]$

example :- $A1 = \text{Harishma}$ $[=UPPER(A1)]$
Result :- HARISHMA.

7) LOWER() :- Converts all text into lower

Syntax :- $[=LOWER(A1)]$

example :- $A1 = \text{HARISHMA}$ $[=LOWER(A1)]$
Result :- harishma

8) TRIM() :- Removes extra space from text.

Syntax :- $[=TRIM(A1)]$

example :- $A1 = \text{Data Analyst}$ $[=TRIM(A1)]$
 ↳ extra space.

Result :- Data Analyst
 ↳ only one space reduced extra space.

9) CONCAT() :-

used to join text from different cells.

Syntax :-

$[=CONCAT(A1, B1)]$

example :-

A
Harishma

B.

Ravichandran

Formula :- $[=CONCAT(A1, B1)]$

Result :- HarishmaRavichandran

↳ no space between.

To add space [= CONCAT(A1, " ", B1)]

⇓

Result :- Harishma Ravichandran

10) TEXT JOIN() :- used to combine text from multiple cell with a separator.

Syntax :- [= TEXT JOIN (separator, ignore empty, text1, text2...)]

Example :-

A	B	C
Harishma	Ravichandran	Bangalore

[= TEXT JOIN (" ", TRUE, A1:C1)]

Result :-

Harishma Ravichandran Bangalore

[" " => helps space between each value]

[TRUE => means to ignore empty cell]

A	B	C
Harishma		Bangalore

[= TEXT JOIN (" ", TRUE, A1:C1)]

Result :-

Harishma Bangalore

11) FIND() :- Find the position of text and its case sensitive.

Syntax :- $[= \text{FIND}("a", A1)]$

Example :- $A1 = \text{Data Analyst}$.

Result :- lower case a is at position 2

12) SEARCH() :- also find the position, but not in case sensitive.

Syntax :-

$[= \text{SEARCH}("A", A1)]$

Example :- $A1 = \text{Data Analyst}$

Result :- 2.

* because search treats "A" and "a" as the same

13) SUBSTITUTE() & REPLACE() :-

* helps for cleaning and correcting data

13) SUBSTITUTE() :- replace specify text with another text.

Syntax :- $[= \text{SUBSTITUTE}(\text{text}, \text{old-text}, \text{new-text})]$

Example :- $A1 = \text{I like Excel}$.

$[= \text{SUBSTITUTE}(A1, "Excel", "SQL")]$

Result :- I like SQL

14) REPLACE() :- replace based on their position

Syntax :-

[= REPLACE (old text, start-num,
num-chars, new text)]

example :- A1 = ABC123.

[= REPLACE (A1, 4, 3, "456")]

Result :- ABC456